An Innovative Hunting-based Quantum-inspired Jaguar Algorithm for Combinatorial Optimization

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***Abstract*—Quantum-inspired optimization (QIO) is a promising technique that simulates quantum behaviors on classical computers to leverage both classical and quantum advantages in tackling complex optimization problems. This study first proposes the quantum-inspired jaguar algorithm (QJA) to solve combinatorial optimization problems (COP). The quantum property helps address COP efficiently when the element increases in the search space. Besides, QJA extracts the crucial concept of the hunting mechanism from the jaguar algorithm (JA), which effectively addresses numerical optimization in classical computing. The featured hunting mechanism enables QJA to automatically assess the algorithm's situation and adaptively update the critical parameters in QIO, thereby performing an efficient search. QJA tackles a complex financial application: portfolio optimization, which requires an effective solution quickly. In the experimental result, this study utilizes real-world financial data to demonstrate the effectiveness of QJA and to compare it with other QIOs, facilitating advancements in quantum finance. The convergence results indicate that QJA can identify higher quality portfolios in the early stages and continue to enhance them throughout the search process. In summary, this study transfers the knowledge and insights gained from the hunting mechanism in classical computing to QIO, integrating both classical and quantum computing techniques to address COP effectively.**

I. Introduction

Quantum devices are promising techniques for tackling intricate, large-scale, real-world applications. However, the current limitations of the universal quantum computer, including scaling size, gate fidelity, and quantum structure, prevent the realization of a fault-tolerant quantum computer. Quantum-inspired optimization (QIO) [1] is a technique for simulating the quantum mechanism in classical computers to obtain a quantum advantage. Quantum behavior, such as the superposition state, offers an advantage over the classical computer in combinatorial optimization problems (COP) when elements are increased in the solution space. Many industrial optimization problems are COP, such as scheduling and portfolio optimization, where portfolio optimization has garnered attention as a realistic and complex real-world application.

To exert the best ability in classical computers, this study learns from an efficient metaheuristic Jaguar algorithm (JA) [2]. JA is a high-efficiency and effective search algorithm in numerical optimization. The core concept of JA is the hunting mechanism to exploit the nearby space to the extreme, which is an efficient method to approach the best solution, as shown in Figure 1. The hunting mechanism first detects a better tendency and then quickly doubles the search tendency until it is not getting better and then exploits the nearby area carefully using a halving method to get accurate historical information.

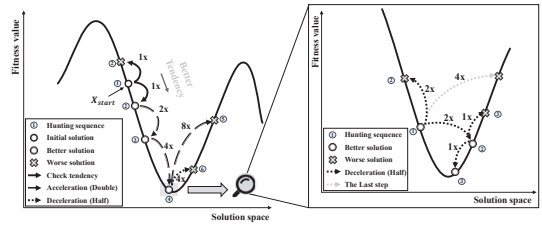

Fig. 1: The hunting mechanism in JA exploits the nearby area along with the tendency efficiently.

This study aims to develop a useful and efficient QIO for COP. Therefore, we extract the hunting search efficiency concept in JA into QIO to propose a hybrid quantum-inspired Jaguar algorithm (QJA), benefiting from the quantum advantage and the classical optimization method while effectively addressing the COP. The proposed hunting-based update mechanism adaptively modifies the magnitude of convergence in the superposition state by identifying the search process trend, thereby enhancing search efficiency. The real-world portfolio optimization is implemented to test the efficacy of QJA with other QIO techniques.

II. Hunting-based Quantum-inspired JA

The proposed QJA extracts the hunting feature of JA, which is to determine the convergence direction with the nearby search area carefully and use the historical information to adaptively expand or narrow down the search space by the power of 2. Within various QIO, the quantum-inspired tabu search algorithm (QTS) [3] is an efficient metaheuristic adapted to solve portfolio optimization effectively [4]. QJA also inherits the core concept of QTS: approaching the best while remaining far from the worst to update the superposition state. The following is the detailed process of QJA:

(i) Initialization: The particle in QJA will be initialized as equal state in superposition as shown in Equation (1).

$$|\Psi\rangle = [\ |\psi_0\rangle\ |\psi_1\rangle\ \cdots\ |\psi_i\rangle\ ],\ where\ |\psi_i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (1)$$

(ii) Measurement: The solution is determined by measuring the superposition with $Pr(|1\rangle) = \langle\psi_i|1\rangle\langle 1|\psi_i\rangle$. If $|\psi_i\rangle = |1\rangle$ means selecting the ith stock into the portfolio, and vice versa.

(iii) Update: The update rule between the global best and local worst solutions (s^{gb} and s^{lw}), determined by the trend ratio [5] as shown in Equation (2), is the core concept for making the superposition converge to the optimal solution, which can accelerate the solution search around the best and avoid the worst area. The rule is shown in Equation (3), where b_i represents the probability of measuring state $|1\rangle$ in bit i.

$$Trend\ Ratio = \frac{Expected\ Return}{Risk} \quad (2)$$

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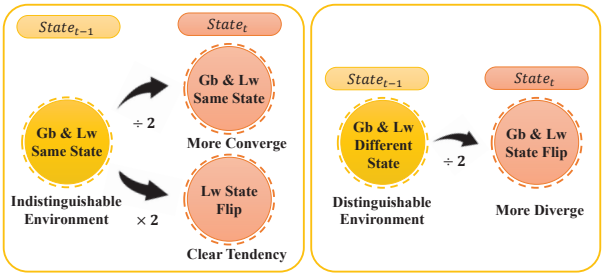

Fig. 2: The adaptive θ adjustment with automatic detection of environment

$$b_i = \begin{cases} b_i + \theta, & if\ s_i^{gb} = 1\ \&\ s_i^{lw} = 0 \\ b_i - \theta, & if\ s_i^{gb} = 0\ \&\ s_i^{lw} = 1 \\ b_i \quad , & if\ s_i^{gb} = s_i^{lw} \end{cases} \tag{3}$$

(iv) Hunting-based adaptive update: QJA procedures focus on adaptively adjusting the superposition state through historical information. The hunting-based adaptive adjustment mechanism is introduced in more detail. The QJA adaptively adjusts the superposition state with a θ. The initial θ is the smallest update volume 2^{-24} according to 0.5. According to the global best and local worst solutions discrepancy with the historical information, the adjustment of θ is determined according to Figure 2. Based on the changes between the previous ($state_{t-1}$) and current ($state_t$) update situation, there are three major cases in Figure 2 to judge the search tendency and then update the magnitude of θ adaptively:

When s_i^{gb} and s_i^{lw} are at the same state at $state_{t-1}$, it means the environment is indistinguishable at that time.

1) Remaining in the same state to converge: If s_i^{gb} and s_i^{lw} states remain in the same condition at $state_t$, it represents that the search tendency might be converged to halve the θ.

2) Change for clear tendency from the same state: If s_i^{lw} flips at $state_t$, it means that the search tendency is clear toward the s_i^{gb} and then double the θ to enhance the convergence.

When s_i^{gb} and s_i^{lw} are in different states at $state_{t-1}$, it means the environment is distinguishable and clear.

3) Change for divergence from a clear tendency: If s_i^{gb} and s_i^{lw} flip, the search environment tendency becomes more divergent and must be checked carefully. The update θ will be halved to slow down the convergence.

III. Experimental Result

The investment target is the top 30 companies based on market value in the Toronto Stock Exchange (TSX). The initial fund is $10 million, and the training-to-testing period is a total of 831, spanning from 2013 to 2022. This study conducts the QJA experiment and compares it with QTS and GQTS, which are the variants of QTS. QJA focuses on using a hunting-based mechanism to adaptively adjust the convergence level. Search efficacy can be improved with a more accurate parameter. In Table I, experimental iterations are 10,000 and 50 independent runs to identify the best solution. The QJA has superior performance over QTS and GQTS in most training-to-testing sliding windows in terms of average solution quality and robustness results. The convergence level can easily recognize the advantage of QJA. It converges to the best solution quickly in an early stage, as shown in Figure 3.

TABLE I: Comparison between QJA and other QIO techniques

Sliding Window [5]		QJA	QTS [3]	Improv.	GQTS [6]	Improv.
M2M	Trend Ratio	**0.649183078**	0.555508996	14.43%	**0.649183078**	0.00%
	#Opt.	**44.15**	1.15	97.40%	2.88	93.48%
M*	Trend Ratio	**0.661772898**	0.569172829	13.99%	**0.661772898**	0.00%
	#Opt.	**44.36**	1.14	97.43%	2.93	93.39%
Q2M	Trend Ratio	**0.226305784**	0.200308551	11.49%	**0.226305784**	0.00%
	#Opt.	**44.02**	1.33	96.98%	2.93	93.34%
Q2Q	Trend Ratio	**0.226678563**	0.200719626	11.45%	**0.226678563**	0.00%
	#Opt.	**43.56**	1.30	97.02%	2.10	95.18%
Q*	Trend Ratio	**0.230733342**	0.204702363	11.28%	**0.230733342**	0.00%
	#Opt.	**43.68**	1.30	97.02%	2.40	94.51%
H2M	Trend Ratio	**0.124955095**	0.109098100	12.69%	0.124893634	0.05%
	#Opt.	**42.31**	1.16	97.26%	3.21	92.41%
H2Q	Trend Ratio	**0.129099089**	0.113266373	12.26%	0.128914708	0.14%
	#Opt.	**41.90**	1.15	97.26%	4.28	89.79%
H2H	Trend Ratio	**0.132441034**	0.115519762	12.78%	0.132072271	0.28%
	#Opt.	**41.50**	1.25	96.99%	3.90	90.60%
H*	Trend Ratio	**0.132907743**	0.115823092	12.85%	0.132538981	0.28%
	#Opt.	**41.00**	1.10	97.32%	3.90	90.49%
Y2M	Trend Ratio	0.070666800	0.063818668	9.69%	0.070676794	-0.01%
	#Opt.	**42.98**	1.56	96.37%	3.17	92.62%
Y2Q	Trend Ratio	**0.073221371**	0.066011769	9.85%	**0.073221371**	0.00%
	#Opt.	**42.33**	1.23	97.09%	3.58	91.54%
Y2H	Trend Ratio	**0.072126131**	0.065830952	8.73%	**0.072126131**	0.00%
	#Opt.	**42.50**	1.25	97.06%	4.45	89.53%
Y2Y	Trend Ratio	**0.065789262**	0.060574832	7.93%	**0.065789262**	0.00%
	#Opt.	**36.73**	1.45	96.05%	7.36	79.96%
831 avg.	Trend Ratio	**0.290984974**	0.252402722	13.26%	0.290950916	0.01%
	#Opt.	**43.20**	1.26	97.08%	3.17	92.66%

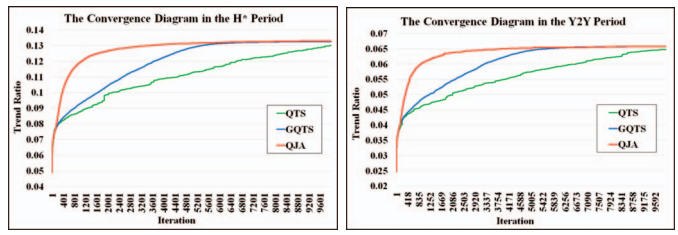

Fig. 3: The convergence levels of QJA are better than other QIO techniques.

IV. Conclusion

This study proposes an innovative QJA that uses JA's efficient hunting optimization ability and adopts it into QIO to enhance search effectiveness. The QJA exerts adaptive environmental detection to adjust the convergence level in the superposition state. The experiment shows that QJA can better address real-world financial portfolio optimization than other QIO techniques and be more efficient. QJA demonstrates a successful example of developing a hybrid quantum-classical optimization that benefits from both classical optimization and quantum behavior advantages.

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